



## Zeyu Wang

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**Work:** Egyetem u. 10 Faculty of Information Technology, 8200 Veszprém (Hungary)

### EDUCATION AND TRAINING

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#### Ph.D. in Biomedical Engineering

*University of Pannonia* [ 1 Sep 2021 – Current ]

City: Veszprém | Country: Hungary | Website: <https://uni-pannon.hu/>

#### Master in Biomedical Engineering

*Shenyang University of Technology* [ 1 Sep 2018 – 29 Jun 2021 ]

City: Shenyang | Country: China | Website: <https://www.sut.edu.cn/index.jsp>

#### Bachelor in Biomedical Engineering

*Shenyang University of Technology* [ 23 Aug 2014 – 17 Jun 2018 ]

City: Shenyang | Country: China | Website: <https://www.sut.edu.cn/index.jsp>

### RESEARCH INTERESTS

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#### Substantive

Cognitive neuroscience, Neurolinguistics, Neuroaesthetics, Epilepsy

#### Methodological:

Neural signal processing, High-performance computing, GPU computing, Machine learning

### PUBLICATIONS

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[2023]

#### [Efficient GPU implementation of the multivariate empirical mode decomposition algorithm](#)

Z. Wang and Z. Juhasz, "Efficient GPU implementation of the multivariate empirical mode decomposition algorithm", *Journal of Computational Science*, vol. 74, p. 102180, Dec. 2023

[2023]

#### [GPU Implementation of the Improved CEEMDAN Algorithm for Fast and Efficient EEG Time-Frequency Analysis](#)

Z. Wang and Z. Juhasz, "GPU Implementation of the Improved CEEMDAN Algorithm for Fast and Efficient EEG Time-Frequency Analysis", *MDPI Sensors (Basel)*, vol. 23, no. 20, p. 8654, Oct. 2023

[2023]

#### [Classifying epileptic phase-amplitude coupling in SEEG using complex-valued convolutional neural network](#)

C. Li, S. Liu, Z. Wang, G. Yuan, "Classifying epileptic phase-amplitude coupling in SEEG using complex-valued convolutional neural network", *Frontiers in Physiology*, vol. 13, p. 1085530, Jan. 2023

[2023]

#### [Massively Parallel EEG Algorithms for Pre-exascale Architectures](#)

Z. Wang and Z. Juhasz, "Massively Parallel EEG Algorithms for Pre-exascale Architectures", European Conference on Parallel Processing (Euro-Par 2023), Limassol, Cyprus, Aug 28 - Sept 1, 2023

[2022]

**[On the Benefits of Empirical Mode Decomposition in Spatio-temporal EEG Analysis](#)**

Z. Wang, Z. Nagy, Z. Juhasz, "On the Benefits of Empirical Mode Decomposition in Spatio-temporal EEG Analysis", 45th Jubilee International Convention on Information, Communication and Electronic Technology, Opatija, Croatia, May 23 - 27, 2022

[2021]

**[Phase-Amplitude Coupling and Epileptogenic Zone Localization of Frontal Epilepsy Based on Intracranial EEG](#)**

H. Ma, Z. Wang, C. Li, J. Chen, Y. Wang, "Phase-Amplitude Coupling and Epileptogenic Zone Localization of Frontal Epilepsy Based on Intracranial EEG", Frontiers in Neurology, vol. 12, p. 718683, Sept 2021

## **HONOURS AND AWARDS**

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[ 1 Sep 2021 ] China Scholarship Council & Hungary Tempus Public Foundation  
**Stipendium Hungaricum Scholarship Program**

[ 5 Jul 2023 ] Euro-Par Foundation  
**Euro-Par 2023 PhD Studentship**

[ 1 Sep 2023 ] Hungary National Research, Development and Innovation Office (NRDI)  
**New National Excellence Programme (ÚNKP)**

[ 23 Nov 2024 ] 2024 International Conference on Science and Engineering of Electronics (ICSEE'2024)  
**Best oral presentation award**

## **CONFERENCES AND SEMINARS**

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[ 22 Nov 2024 – 24 Nov 2024 ] Wuhan, China  
**2024 International Conference on Science and Engineering of Electronics (ICSEE 2024)**  
Oral presentation

[ 25 Jun 2024 – 29 Jun 2024 ] Vienna, Austria  
**2024 Federation of European Neuroscience Societies Forum (FENS 2024)**  
Poster presentation

[ 25 Jan 2024 – 26 Jan 2024 ] Pécs, Hungary  
**2024 International Neuroscience Conference (INC 2024)**  
Poster presentation

[ 28 Aug 2023 – 1 Sep 2023 ] Limassol, Cyprus  
**29th International European Conference on Parallel and Distributed Computing (Euro-Par 2023)**  
Oral presentation

[ 23 May 2022 – 27 May 2022 ] Opatija  
**45th Jubilee International Convention on Information, Communication and Electronic Technology (IEEE-MIPRO 2022)**  
Oral presentation

[ 20 Jun 2022 – 31 May 2024 ] Budapest, Hungary

**12th, 13th, and 14th GPU Day International Conference**

Oral presentation

[ 3 Dec 2021 – 7 Dec 2024 ] Veszprém & Szeged, Hungary

**XXXIV, XXXV, XXXVI, XXXVII. Neumann Colloquium**

Oral presentation

[ 20 Jul 2020 – 24 Jul 2020 ] Montreal, Canada

**42nd Annual International Conferences of the IEEE Engineering in Medicine (IEEE-EMBC 2020)**

Oral presentation

[ 24 Aug 2018 – 27 Aug 2018 ] Shenyang

**2018 IEEE International Conference on Intelligence and Safety for Robotics (IEEE-ISR 2018)**

Oral presentation

## NETWORKS AND MEMBERSHIPS

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[ 7 Feb 2024 – Current ] Brussels

**Member of Federation of European Neuroscience Societies**

Link: <https://www.fens.org/>

[ 31 Dec 2023 – Current ] Budapest

**Member of Hungarian Neuroscience Society (MITT)**

Link: <https://www.mitt.hu/en>

[ 3 Dec 2024 – Current ] Budapest

**Member of John von Neumann Computer Society (NJSZT)**

## TEACHING EXPERIENCE

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**Faculty of Information Technology, University of Pannonia**

Teaching Assistant to Dr. Zoltan Juhasz

- GPU Programming 2023/24/1
- GPU Programming 2024/25/1